

of monitoring and observes additional parameters like oxygen saturation in the body. Unlike traditional systems, these trends are based on clinical protocols designed by expert neonatologists, which are out for comparison by the prediction engine.

The solution also aims to solve the key pain points by potentially predicting life-threatening infections based on the variation in vital parameters. Remote consultation specialists now have access to a multitude of information and complete profile of the patient, while making it convenient to diagnose and delivering the right care at the right time.

With the IoT, constrains of distance, time, better predictability and manual intervention disappear. This online saviour or connected ICUs leverage Cloud, intelligent algorithms, prediction engines and medical monitoring devices present near the patient to gather vital data, while replacing a designated person providing personalised care.

Connecting these care units

Breaking the barriers of non-connectivity, ICU Connect and NICU Connect systems leverage the power of real-time monitoring of vital health parameters, leveraging Wi-Fi, LAN and mobile networks.

Real-time or historical vital stats could be viewed at the nursing station, physician's office, central monitoring station or from any distance using a smartphone application.

Additionally, the system can be used to program customised alerts with respective health parameters and conditions that flag up in case of deviations from set benchmark levels. Real-time notifications are sent to the concerned neonatologists and intensivists anywhere in the world. This could prevent neonates or very sick people going into crisis mode.

Cloud connectivity

ICU Connect platform is built on open source technologies to control the cost

of solutions. This Cloud based solution needs minimum infrastructure at minimum cost of deployment.

ICU Connect and NICU Connect solutions come with three main sub-systems that include both hardware and software components.

Aggregator connectivity hub.

This hardware hub has multi ports, which connect to multiple devices. Data streams into the Cloud or an external device with the help of Wi-Fi or LAN. Connectivity hub also has data buffering capabilities to store required minimum data before sending it to the Cloud.

ICU Connect and NICU Connect servers.

Data aggregation from different devices is implemented in this stage. Data-listening mechanism, device driver implementation, HL7 conversion and data-publishing mechanism help accumulate and process data. Algorithms for trending critical diseases are the key components of the application.

ICU Connect and NICU Connect client applications.

System components at this stage include real-time data broadcaster and message subscriber. Web socket protocols are implemented for effective browser based readings on the computer, tablet or smartphone. At the client end, reporting systems include plotting and charting mechanisms to assist with trends and patterns that again need expert eyes for evaluation.

Similarities and differences

Most competitor applications are integrated applications for electronic medical records or hospital information systems that require expensive IT infrastructure, making it a costly affair for its beneficiaries. "The only solution was to use open source software and indigenous connectivity engines with inbuilt HL7 conversion engines," says Kurady.

HL7 message helps manage medical records by transmitting new or updated documents, or by transmitting important status

information and/or updates for the record. Real-time monitoring through analytics, prediction engines combined with HL7 conversions could be viewed and analysed using any smartdevice. Customised alerts could be set to capture the attention of the expert or doctor and can be set by the expert himself.

Crisis prediction becomes achievable by leveraging data from every component available on the bedside of the patient such as patient monitor, ventilator, warmers and even syringe pumps. This implies that vital information could be extracted from these systems as well.

The future of critical healthcare

As vital health data starts pouring in from the machines through wireless or wired media to the aggregator device, the analytics platform powered by prediction algorithm kicks in to create trends and real-time charts for reporting.

Statics and readings could be logged in anywhere from a stationary or a moving infrastructure, which is an add-on. This approves the fact that ICU Connect and NICU Connect can be installed and connected anywhere in a cost-effective manner, fulfilling a healthy offering for a critically-ill patient.

"In essence, we are offering uberisation of medical device data, connecting the doctor to the critical patient's data via universal critical care real-time remote monitoring system with customisable alerts and predictive algorithms, enabling healthcare delivery beyond boundaries," says Dr Prasanna Ganapa, co-founder and head of strategy, RTWO Healthcare Solutions.

In a never-concluding note, Cloud based healthcare is truly catching up and essentially will leverage all computing power that there is using the IoT. ICU Connect platform is on constant upgradation with respect to clinical features, aggregator models and healthcare analytics. **EFY**